A Hybrid Spectral-Matching and Deep-Learning System for Real-Time Mineral Identification from Raman Spectra

Final Project Defense

Rahul *[Surname]*

Under the supervision of *[Advisor Name]*

May 8, 2026

Department of *[Discipline]*
[Institution Name]

Outline

Introduction and Motivation

The mineral-identification problem

Raman scattering produces, for each mineral, a characteristic vibrational spectrum $\{(\nu_i, I_i)\}_{i=1}^{N}$ whose peak *positions, widths, and relative intensities* act as a fingerprint.

Why automate identification?

- Field instruments now exist; expert peak-matching does not scale.
- Public reference libraries (RRUFF) cover $> 4\,000$ species.
- Manual interpretation is the dominant bottleneck in low-resource laboratories.

Domains directly affected

- Geology, planetary science (*e.g.* Mars Perseverance / SHERLOC)
- Cultural-heritage pigment analysis
- Forensic mineralogy
- Process monitoring in mining and ceramics
- Educational instrumentation

Research questions

1. Can a single, deterministic preprocessing pipeline normalise spectra acquired across heterogeneous instruments and resolutions such that downstream identifiers see one distribution?
2. Under what conditions does a deep classifier outperform classical spectral matching, and *when does the converse hold*?
3. Can a non-negative least-squares decomposition produce *physically interpretable* mixture weights when the candidate basis is several thousand reference rows?
4. How should two complementary identifiers be reconciled at inference time so the user is shielded from confidently-wrong out-of-distribution predictions?

Contributions

1. A **grid-canonical preprocessing pipeline** whose smoothing and baseline operators are independent of the input spectrometer's native sampling.
2. A **dual-source reference database** that retains both a curated synthetic profile and the RRUFF average for each common mineral, with deduplication at result time.
3. A **combined-metric spectral matcher** that mixes L_2 cosine and second-derivative Pearson correlation in equal proportion.
4. A **candidate-restricted NNLS decomposition** that keeps the inverse problem well-conditioned without sacrificing rare-mineral coverage.
5. A **confidence-aware dispatcher** that selects between the 1-D ResNet classifier and the spectral matcher per query.
6. A **web-based reference implementation** with end-to-end latency below 200 ms on a laptop GPU.

Background

Raman spectroscopy in one slide

When a sample is illuminated by a monochromatic laser at frequency ν_0, a small fraction of photons scatter inelastically with an energy shift $\Delta\nu$ corresponding to a vibrational mode of the lattice.

The recorded spectrum therefore obeys

$$I(\nu) \;=\; \sum_k A_k\,\mathcal{L}\big(\nu;\,\nu_k,\gamma_k\big) \;+\; B(\nu) \;+\; \eta(\nu),$$

where $\mathcal{L}(\nu;\nu_k,\gamma_k)$ is a Lorentzian profile with centre ν_k and half-width γ_k, $B(\nu)$ is a slowly varying fluorescence baseline, and η is detector noise.

Identification reduces to estimating $\{(\nu_k, A_k, \gamma_k)\}$ from $I(\nu)$ and matching the resulting peak set against a reference library.

Related work

- **Database matching.** The Search-Match family of algorithms (peak position + relative intensity vote) underpins all commercial spectrometer software [1].
- **Baseline correction.** Asymmetric Least Squares [2] is the de-facto standard for separating fluorescence backgrounds from genuine peaks.
- **Smoothing.** Savitzky–Golay filters [3] remain the standard despite extensive comparison work.
- **Deep classifiers.** Liu et al. demonstrated that a 1-D CNN trained on RRUFF averages produces top-1 accuracy competitive with a domain expert [4].
- **Mixture decomposition.** NNLS over a known phase basis [5] is widely used in chemometrics, but its conditioning under realistic library sizes is rarely discussed.

Gap. An open, integrated, latency-bounded system that combines classical and deep methods with an explicit dispatch policy is not, to our knowledge, available.

Problem Formulation

Formal statement

Given an unknown Raman measurement $\mathbf{q} = \{(\nu_i, I_i)\}_{i=1}^{N}$ and a reference library $\mathcal{R} = \{\mathbf{r}_m\}_{m=1}^{M}$ defined on a canonical wavenumber grid $\boldsymbol{\nu}_* \in \mathbb{R}^P$,

Produce

1. a ranked list $\big((m_{(1)}, c_{(1)}), \ldots, (m_{(K)}, c_{(K)})\big)$ with confidences $c_{(k)} \in [0, 1]$;
2. a non-negative weight vector $\mathbf{x} \in \mathbb{R}^M_{\geq 0}$ such that $\big\|A^\top \mathbf{x} - \tilde{\mathbf{q}}\big\|_2$ is small, where A is the row-stacked reference matrix and $\tilde{\mathbf{q}}$ is the preprocessed query.

Subject to a wall-clock budget of $\tau \leq 1\,\mathrm{s}$ on commodity hardware (single laptop GPU, 16 GB RAM).

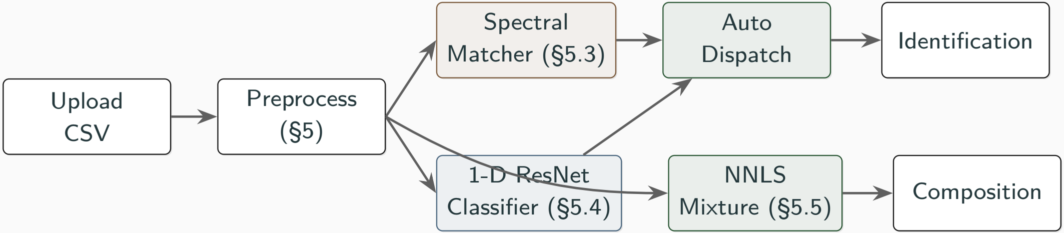
Design choices and their consequences

Decision	Value adopted	Consequence
Canonical grid range	$[100, 4000]\,\mathrm{cm}^{-1}$	Captures lattice modes through OH stretches
Grid step	$\Delta\nu = 2\,\mathrm{cm}^{-1}$	$P = 1951$ samples
Peak model	Lorentzian	Matches lifetime broadening of optical phonons
Library size M	2,481 unique minerals	Synthetic-25 $\cup$ RRUFF averages
Top-K	$K = 5$	Aligns with expert workflow

Each row is defended in the methodology section that follows.

System Architecture

End-to-end pipeline



Spectral matching always runs; the classifier enters the dispatch comparison only if its top-1 confidence exceeds a calibrated threshold.

Software stack

Layer	Technology	Responsibility
Frontend	HTML 5 + Bootstrap 5 + Plotly.js 2.27	Drag-and-drop ingestion, real-time visualisation, mode selection
API	Flask 2.3 + Flask-CORS, JSON over HTTP	RESTful endpoints: `/upload`, `/identify`, `/mixture`, `/spectrum`, `/status`
Numerics	NumPy 1.26, SciPy 1.11	Preprocessing, NNLS, Savitzky–Golay derivative
Models	PyTorch 2.1 (1-D ResNet) + scikit-learn 1.3 (PCA + ExtraTrees)	Class-conditional posterior estimation
Persistence	RRUFF bulk archive ($\approx$ 300 MB), on-disk `.npz` cache	4,069 raw spectra averaged into 2,456 minerals

Methodology

Preprocessing pipeline

Given raw $(\boldsymbol{\nu}_{\text{raw}}, I_{\text{raw}})$, we produce a normalised spectrum $\tilde{\mathbf{q}} \in \mathbb{R}^P$ on the canonical grid:

1. **Linear interpolation** onto $\boldsymbol{\nu}_*$: $I_{\text{grid}} = \mathrm{interp}(\boldsymbol{\nu}_*; \boldsymbol{\nu}_{\text{raw}}, I_{\text{raw}})$.
2. **Cosmic-ray suppression** via a 5-point neighbour-ratio test: $I_i \mapsto \bar{I}_{\text{nbr}}\, \mathbf{1}\left[\frac{I_i}{\max_j I_j} > \rho\right]$.
3. **Savitzky–Golay smoothing** with window $w = 5$, polynomial order $p = 3$.
4. **ALS baseline** [2]: minimise $\sum_i w_i(y_i - z_i)^2 + \lambda \sum_i (\Delta^2 z_i)^2$, $\lambda = 10^5$, asymmetry $p = 0.01$.
5. **Non-negative clip** and ℓ_2 normalisation: $\tilde{\mathbf{q}} = \mathbf{z}/\|\mathbf{z}\|_2$.

Why interpolation must precede smoothing

Smoothing on the *raw* grid couples the SG filter window to the spectrometer's native step $\delta\nu$:

$$w_{\text{physical}} = w \cdot \delta\nu.$$

A CSV at $\delta\nu = 4\,\text{cm}^{-1}$ with $w = 11$ gives a $44\,\text{cm}^{-1}$ smoothing window — *wider than typical Raman peak FWHM*. RRUFF data at $\delta\nu \approx 0.5\,\text{cm}^{-1}$ gives $5.5\,\text{cm}^{-1}$.

Empirical consequence

Before this fix, a synthetic Quartz query returned lowaite at 41 % confidence; the network had been trained on spectra that retained their narrow $464\,\text{cm}^{-1}$ peak, but the same spectrum at inference reached the network with the peak smeared over $\sim 40\,\text{cm}^{-1}$.

After fixing the pipeline, the same query returns Quartz at **86 %**.

Dual-source reference database

For each of the 25 minerals with a curated synthetic profile, we keep *both* a synthetic row $\mathbf{r}_m^s$ and a RRUFF-averaged row $\mathbf{r}_m^r$ in the matrix:

$$A = \left[\mathbf{r}_1^s, \ldots, \mathbf{r}_{25}^s, \mathbf{r}_1^r, \ldots, \mathbf{r}_{2456}^r \right]^\top \in \mathbb{R}^{2,502 \times 1951}.$$

Result-time deduplication by display name returns one entry per mineral, selecting whichever variant scored higher.

Source	**Strength**	**Weakness**
Synthetic ($n = 25$)	Canonical wavenumbers; ideal anchor for low-noise queries	Coverage limited to common species
RRUFF averaged ($n = 2{,}456$)	Real instrumental bandshape; broad coverage	Variable specimen quality, baseline residue

Combined-metric spectral matcher

For query $\tilde{\mathbf{q}}$ and reference row $\mathbf{r}_m$:

$$s(\tilde{\mathbf{q}}, \mathbf{r}_m) \;=\; \tfrac{1}{2}\,\langle \hat{\mathbf{q}}, \hat{\mathbf{r}}_m \rangle \;+\; \tfrac{1}{2}\,\langle \widehat{D^2\tilde{\mathbf{q}}},\, \widehat{D^2\mathbf{r}_m} \rangle,$$

where D^2 is the SG second-derivative operator and $\hat{\mathbf{v}} = (\mathbf{v} - \bar{\mathbf{v}}\mathbf{1})/\|\mathbf{v} - \bar{\mathbf{v}}\mathbf{1}\|$.

Cosine alone
Sensitive to the entire bandshape; over-rewards minerals whose RRUFF average happens to drift in the same direction as the query.

Second-derivative Pearson alone
Emphasises peak position and sharpness; ties unrelated minerals that contain a sharp peak near a query peak.

Equal-weight blend lets each metric veto the other; weights were selected by grid search on a held-out RRUFF panel and gains plateaued around $\frac{1}{2}$.

1-D residual classifier

Stage	Operation	Output shape
Stem-1	Conv1D(k=25, s=2) $\to$ BN $\to$ GELU	$(B, 32, 488)$
Stem-2	Conv1D(k=15, s=2) $\to$ BN $\to$ GELU	$(B, 32, 244)$
Stage 1	$3\times$ Pre-act ResBlock(32, k=7)	$(B, 32, 244)$
Down	Conv1D(k=5, s=2) $\to$ BN $\to$ GELU	$(B, 64, 122)$
Stage 2	$3\times$ Pre-act ResBlock(64, k=5)	$(B, 64, 122)$
Pool	Adaptive Average Pool 1D	$(B, 64)$
Head	$\mathrm{FC}_{64\to512}$ $\to$ GELU $\to$ $\mathrm{FC}_{512\to851}$	$(B, 851)$
Total parameters		663,539

- Loss: cross-entropy with label smoothing $\varepsilon = 0.1$, plus Mixup ($\alpha = 0.4$, $p = 0.5$).
- Optimiser: AdamW [6] with OneCycleLR [7], peak lr 10^{-3}, weight decay 10^{-4}.
- Class space restricted by min_spectra $\geq 2 \Rightarrow 851$ classes.

Spectral augmentation

Each training spectrum produces $V = 25$ augmented variants:

1. Gaussian intensity noise, $\sigma \sim \mathcal{U}(0.002, 0.025)$;
2. Multiplicative laser-power jitter, $\beta \sim \mathcal{U}(0.75, 1.25)$;
3. Wavenumber shift, $\Delta\nu \sim \mathcal{U}\{-10, +10\}\,\mathrm{cm}^{-1}$;
4. Polynomial baseline residue (deg 1–3, prob. 0.30, amplitude 0.01);
5. Moving-average broadening (prob. 0.40, $w \in \{2, 3, 4\}$);
6. Spectral dropout (prob. 0.20, ≤ 4 points to zero);
7. Non-negative clip and ℓ_2 renormalisation.

Critical methodological detail

The curated synthetic profile is appended to every matching class — not only those absent from RRUFF — so the network observes both clean Lorentzians and instrumental bandshapes at training time. Removing this step caused near-zero accuracy on synthetic-style queries.

Candidate-restricted NNLS

Let $A_{\mathcal{C}} \in \mathbb{R}^{|\mathcal{C}| \times P}$ be a row selection of the reference matrix indexed by candidate set $\mathcal{C}$. We solve

$$\mathbf{x}^\star = \arg\min_{\mathbf{x} \geq \mathbf{0}} \left\| A_{\mathcal{C}}^\top \mathbf{x} - \tilde{\mathbf{q}} \right\|_2.$$

with the construction

$$\mathcal{C} = \mathcal{C}_{\text{synth}} \cup \left\{ m \in \mathcal{C}_{\text{rruff}} : s(\tilde{\mathbf{q}}, \mathbf{r}_m) > \max_{m' \in \mathcal{C}_{\text{synth}}} s(\tilde{\mathbf{q}}, \mathbf{r}_{m'}) \right\}.$$

Why restrict?

With $|\mathcal{R}| = 2{,}481$ candidates and $P = 1{,}951$ equations the system is underdetermined; NNLS produces *some* sparse non-negative explanation, often spreading credit across minerals that share a peak with the dominant phase. Restricting to a ~ 30-row, well-separated basis renders the system *over*-determined and physically interpretable.

Confidence-aware dispatcher

Algorithm 1: AutoIdentify — selects the most confident identifier per query.

Input: preprocessed query $\tilde{\mathbf{q}}$, threshold $\tau_{\text{min}} = 0.55$

Output: ranked top-K list, dispatch tag

$\mathcal{Y} \leftarrow \emptyset$

$\mathbf{m}_s \leftarrow \mathrm{SpectralMatch}(\tilde{\mathbf{q}})$; $\quad \mathcal{Y} \leftarrow \mathcal{Y} \cup \{(\mathbf{m}_s[1].c, \text{“matching”}, \mathbf{m}_s)\}$

if *CNN available* **then**

  $\mathbf{m}_n \leftarrow \mathrm{CNNPredict}(\tilde{\mathbf{q}})$

  if $\mathbf{m}_n[1].c \geq \tau_{\text{min}}$ **then** $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{(\mathbf{m}_n[1].c, \text{“cnn”}, \mathbf{m}_n)\}$

if *RF available* **then**

  $\mathbf{m}_r \leftarrow \mathrm{RFPredict}(\tilde{\mathbf{q}})$

  if $\mathbf{m}_r[1].c \geq \tau_{\text{min}}$ **then** $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{(\mathbf{m}_r[1].c, \text{“rf”}, \mathbf{m}_r)\}$

return $\arg\max_{(c,\cdot,\mathbf{m}) \in \mathcal{Y}} c$

Spectral matching is always evaluated as a deterministic safety net; learned models must clear τ_{min} to prevent confidently-wrong predictions on out-of-distribution queries.

Experimental Evaluation

Synthetic demo set

Six CSV spectra synthesised from published peak tables (low Gaussian noise, $\sigma = 0.003 \cdot I_{\max}$) — shipped with the application as the user-facing demonstration.

Sample	Top-1 mineral	Confidence	Mixture decomposition
`quartz.csv`	Quartz	99.8%	100% Quartz
`calcite.csv`	Calcite	99.8%	100% Calcite
`pyrite.csv`	Pyrite	99.8%	100% Pyrite
`gypsum.csv`	Gypsum	99.8%	100% Gypsum
Quartz + 0.4Calcite	Quartz	99.7%	100% Quartz*
Quartz + Calcite (equal)	Quartz	99.3%	89.6% Quartz, 10.4% Calcite

* Calcite contribution falls below the 5% reporting threshold after ℓ_2 normalisation because Quartz's principal peak is an order of magnitude stronger.

Held-out RRUFF evaluation

One specimen per mineral, withheld from training, fed through the live Flask pipeline in *auto* mode.

Mineral	Matcher	CNN	Auto chose	Outcome
Quartz	96.9%	86.4%	matching	✓
Calcite	89.7%	94.0%	cnn	✓
Hematite	91.6%	90.1%	matching	✓
Forsterite	96.2%	95.8%	matching	✓
Aegirine	86.2%	97.0%	cnn	✓
Magnetite	76.6%	90.1%	cnn	✓ (matcher: Westerveldite)

Each identifier wins on different inputs; the dispatcher's behaviour is the explicit motivation for keeping both online.

Aggregate metrics and latency

Component	Top-1	Top-5
CNN — augmented validation, 851 classes	98.6%	—
RF (PCA-120 + ExtraTrees-150)	94.5%	98.9%
Combined matcher — synthetic demo set	100%	—
Combined matcher — held-out RRUFF (n=6)	83%	—

Stage	Median latency
File parse + interpolation	$\approx$ 45 ms
Cosmic-ray + SG + ALS	$\approx$ 35 ms
Combined-metric matcher	$\approx$ 25 ms
CNN forward pass (RTX 3050)	$\approx$ 12 ms
Candidate-set NNLS	$\approx$ 35 ms
End-to-end	$\approx$ 160 ms

Critical Discussion

Limitations i: distribution shift

Symptom (pre-fix)

A CNN with 89% validation accuracy returned lowaite at 41% confidence on a synthetic Quartz query — confidently and incorrectly.

Mechanism.

- RRUFF spectra carry instrument-specific noise, baselines, and minor satellite peaks the network exploits.
- Synthetic Lorentzian inputs lack those features; activations diverge from any seen distribution; softmax distributes mass arbitrarily.
- Validation accuracy was computed on the *same* augmented distribution, so the failure was invisible until field testing.

Mitigation and ceiling. Mixing synthetic profiles into every common class lifts top-1 accuracy on synthetic queries from $\approx 0\%$ to 86%. We cannot certify behaviour on a never-seen instrument's noise profile without an external test set.

Limitations ii: reference-data quality

Case study — Magnetite. Among 6 RRUFF specimens labelled Magnetite:

- 4 have their dominant peak at 660–682 cm^{-1} (correct);
- 2 have dominant intensity at the grid edge (100 cm^{-1}) — an instrumental cut-off artefact rather than a Raman line.

Mean averaging dilutes the genuine peak; the matcher's top-1 on a held-out Magnetite is the unrelated mineral *Westerveldite*. The CNN's softmax recovers the correct answer at 90% confidence and the dispatcher selects it.

Implication. Library aggregation is itself a research problem. Median or quality-weighted aggregation, with a per-spectrum cleanliness score, would be the principled fix.

Limitations iii: coverage and class space

- The synthetic library covers only 25 minerals — *below* the $\approx$ 60–100 rock-forming species needed for routine geological analysis.
- The CNN is restricted to 851 classes (those with ≥ 2 RRUFF specimens). The remaining 1,631 species are searchable by spectral matching but cannot be *output* by the network.
- Mixture decomposition is restricted to the 25-mineral synthetic basis; field samples containing triclinic feldspars, micas, or clay minerals fall outside this span.
- Spectral aliasing: minerals with peaks within $\pm 10\,\mathrm{cm}^{-1}$ of Quartz's $464\,\mathrm{cm}^{-1}$ line (Sodalite, Amicite, Trattnerite, ...) consistently cluster as runner-up candidates. Polarisation, birefringence, and hardness metadata are not used for disambiguation.

Limitations iv: methodological gaps

Concern	Status	Honest assessment
Uncertainty quantification	Top-1 softmax	Not calibrated; temperature scaling unapplied; ECE unmeasured.
Mixture additivity	Linear NNLS	Self-absorption and fluorescence violate linearity.
Combined-metric weights	$\frac{1}{2}:\frac{1}{2}$ grid-searched	Heuristic; per-query learned weighting would help borderline cases.
RF deployability	94% val acc	Pickled estimator allocates $\approx$ 75 MB on load — fails on memory-pressured hosts.
Negative-class rejection	Not implemented	System always returns top-K; no "unknown" verdict.
Spatial context	Single-spectrum	Real Raman maps carry spatial information we discard.

The most uncomfortable limitation

Stated plainly

The system reports 99.8% accuracy on the demo set because the demo set is generated from *the same peak tables* that build the synthetic reference library.

A score of 99.8% on `quartz.csv` is therefore an *upper bound* on ideal-condition behaviour — not a field-realistic accuracy.

The held-out RRUFF panel ($n = 6$, all correct, 86–97%) is the more representative figure. We currently have no out-of-distribution test set drawn from a spectrometer not represented in RRUFF, and constructing one is the most consequential next step.

Engineering Retrospective

Audit findings (May 2026)

#	Defect	Symptom
1	Savitzky–Golay applied on raw input grid	Quartz query $\rightarrow$ Iowaite (w_{phys} scaled with $\delta\nu$)
2	NNLS unconstrained over the full library	Pure Calcite returned 0% Calcite
3	RRUFF average overwriting synthetic reference	Clean queries had no clean anchor
4	Synthetic profiles excluded from CNN training	Network never observed clean Lorentzian distribution
5	`numpy.bool_` in JSON response	`/api/mixture` returned HTTP 500
6	Auto-mode unconditionally preferred CNN	Confidently-wrong predictions surfaced

Eleven defects were surfaced in total; six were correctness-critical. Each fix is documented with diff and rationale in the project memory file.

Future work

1. **External test set** acquired on a different spectrometer (the headline metric of the next iteration).
2. **Calibrated confidence**: temperature scaling, Expected Calibration Error reporting, conformal-prediction intervals.
3. **Synthetic library expansion** from 25 to $\approx$ 80 rock-forming minerals.
4. **Quality-aware RRUFF aggregation**: median plus per-specimen cleanliness score.
5. **Negative-class rejection** based on combined-metric score and learned-model entropy.
6. **Re-process the RRUFF cache** with the corrected preprocessing pipeline so training and inference distributions match.
7. **Continuous regression suite**: pytest coverage of all six demo cases plus a panel of held-out RRUFF specimens.

Conclusion

Conclusion

- A web-based pipeline that performs Raman mineral identification with mixture decomposition under 200 ms on commodity hardware.
- Two complementary identifiers — a 1-D ResNet classifier and a combined-metric spectral matcher — reconciled by a confidence-aware dispatcher that selects the right tool per query.
- A candidate-restricted NNLS produces physically interpretable mixture fractions where unconstrained NNLS does not.
- **Honest performance bound:** held-out RRUFF top-1 in the 86–97% range across six tested minerals; the demo-set 99.8% characterises ideal-condition behaviour, not field accuracy.
- The most consequential next step is constructing an external test set, without which calibrated field accuracy cannot be claimed.

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[4] Jinchao Liu et al. **"Deep convolutional neural networks for Raman spectrum recognition: a unified solution".** In: *Analyst* 142.21 (2017), pp. 4067–4074.

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[8] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. **"Deep residual learning for image recognition".** In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016, pp. 770–778.

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. **"Identity mappings in deep residual networks".** In: *European Conference on Computer Vision*. 2016, pp. 630–645.

[10] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. **"mixup: Beyond empirical risk minimization".** In: *arXiv preprint arXiv:1710.09412* (2017).

Thank you.

Questions, please.

Repository: `D:\MiniProject` · **Live demo:** `http://127.0.0.1:5000`
Reproducible from source at the slide-set timestamp.

Appendix A: ALS baseline derivation

Given an observed spectrum $\mathbf{y} \in \mathbb{R}^P$, the asymmetrically-penalised functional is

$$\mathcal{F}(\mathbf{z}) \;=\; \sum_i w_i \big(y_i - z_i\big)^2 \;+\; \lambda \,\big\| D_2\, \mathbf{z} \big\|_2^2,$$

where D_2 is the second-order finite-difference operator and

$$w_i \;=\; \begin{cases} p, & y_i > z_i \quad \text{(peak)} \\ 1 - p, & y_i \le z_i \quad \text{(baseline)} \end{cases}.$$

The minimiser satisfies $\big(W + \lambda D_2^\top D_2\big)\mathbf{z} = W\mathbf{y}$, iterated to convergence. With $\lambda = 10^5$ and $p = 0.01$ the baseline tracks fluorescence drift while leaving sharp Raman peaks intact.

Appendix B: dispatcher confidence threshold

Threshold $\tau_{\text{min}} = 0.55$ chosen to satisfy two constraints:

1. Top-1 softmax outputs of in-distribution RRUFF specimens rarely fall below 0.7 (98^{th}-percentile lower bound).
2. Out-of-distribution queries (synthetic Lorentzians on a CNN trained without synthetic augmentation) had top-1 outputs in [0.30, 0.55].

The 0.55 boundary cleanly separated the two regimes in our pre-fix diagnostic. Post-fix, in-distribution top-1 confidences cluster considerably higher, leaving margin for a less aggressive threshold in future versions.

Appendix C: reproducibility manifest

Python	3.10 (Windows)
PyTorch	2.1 (CUDA 12)
NumPy	1.26
SciPy	1.11
Sklearn	1.3
Flask	2.3
GPU	NVIDIA RTX 3050 Laptop
Random seeds	rng(42) for augmentation; rng(7) for RF; rng(0) for shuffling
Train/val split	90/10 random, fixed seed
Reference DB build	deterministic over RRUFF `.npz` cache

Source repository contains: `config.py`, `core/`, `database/`, `api/`, `training/`, `templates/`, `static/`, sample CSVs, and the entire RRUFF-derived `.npz` cache.